

Ryan Cho

Full-Stack Software Engineer — Python & Distributed Platforms

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PROFESSIONAL SUMMARY

Full-stack software engineer with 8+ years delivering production Python applications, APIs, and data-driven web platforms in regulated finance environments. Skilled in FastAPI service design, asynchronous ETL workflows, PostgreSQL data modeling, and TypeScript/Vue frontends. Deployed containerized systems on Kubernetes using Docker and Helm while collaborating with product and analytics stakeholders on cross-functional delivery.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and maintained **production Python services** powering portfolio analytics workflows, exposing **RESTful APIs** consumed by internal traders and operations teams across multiple business units.
- Built **FastAPI** endpoints and asynchronous workers to process high-volume financial data feeds, reducing end-to-end pipeline latency through **event-driven** task orchestration and connection pooling.
- Designed **TypeScript/Vue** dashboards for investment research and reporting, partnering with **product managers** to translate stakeholder requirements into responsive, data-dense interfaces.
- Implemented **ETL pipelines** ingesting market and reference data into **PostgreSQL**, applying schema migrations and indexing strategies that improved query performance for analytics workloads.
- Containerized microservices with **Docker** and deployed releases via **Helm charts** to **Kubernetes** clusters, standardizing environment configuration across staging and production.
- Integrated **LLM-powered automation** prototypes for document summarization and workflow assistance, collaborating with analytics engineers to evaluate orchestration patterns before production rollout.
- Established **CI/CD** gates and observability hooks for backend services, enabling faster release cycles while maintaining audit-friendly deployment logs required in regulated environments.
- Mentored junior engineers on **API design**, testing practices, and code review standards, contributing to team-wide engineering guidelines adopted across two product squads.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Python** tooling and services supporting trade operations and risk reporting, delivering reliable backend components within a globally distributed engineering organization.
- Authored **SQL** queries and stored procedures against **relational databases** to power daily reconciliation dashboards used by technology and business analysts.
- Built internal **web APIs** connecting front-office applications to centralized data stores, enforcing authentication and input validation consistent with enterprise security policies.
- Supported **message-oriented** integration between legacy mainframe systems and modern services, helping migrate batch workflows toward more resilient asynchronous processing.
- Partnered with **data scientists** and quantitative analysts to productionize research models, packaging notebooks into maintainable service endpoints with scheduled refresh jobs.
- Automated deployment packaging with **Docker** images for sandbox environments, reducing manual setup time for cross-team integration testing.
- Participated in **agile** delivery ceremonies and technical design reviews, communicating trade-offs clearly to product owners and senior stakeholders during quarterly planning.

Software Engineering Intern Chexology

Jun/2018 - May/2019

Remote | 17th Floor, New York

- Contributed to **full-stack** features for a restaurant technology platform, implementing backend endpoints and UI components used by venue operators in production.
- Wrote **Python** scripts and API integrations supporting check-in and payment flows, applying unit tests to guard against regressions in high-traffic weekend periods.
- Developed **JavaScript** interfaces for administrative tooling, improving form validation and error handling based on feedback from customer success teams.
- Queried **PostgreSQL** schemas to support reporting features, optimizing slow lookups with targeted indexes after profiling ORM-generated statements.
- Collaborated with senior engineers during **code reviews**, learning service decomposition patterns and async job handling used across the core platform.
- Documented local development setup and API contracts, helping onboard subsequent interns and reducing environment-configuration support requests.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web pages** for a nonprofit foundation site, translating design mockups into semantic HTML, CSS, and lightweight JavaScript interactions.
- Integrated **third-party APIs** for newsletter signup and donation flows, validating payloads server-side to prevent malformed submissions.
- Optimized page load performance through asset minification and image compression, improving **Lighthouse** scores ahead of a public fundraising campaign launch.
- Coordinated with remote stakeholders on content updates and release timing, ensuring deployments aligned with marketing milestones despite a compressed project timeline.
- Maintained version-controlled source and deployment checklists, establishing early habits around **git workflow** and environment promotion reused throughout later roles.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, Vue.js, JavaScript, HTML5, CSS3, Responsive UI Design

Backend: Python, FastAPI, REST APIs, Asynchronous Services, Event-Driven Architecture, ETL Pipelines

Databases: PostgreSQL, SQL, SQLAlchemy, Relational Data Modeling, Snowflake

Cloud & DevOps: Docker, Kubernetes, Helm, Terraform, CI/CD, Container Orchestration

Monitoring & Tools: Git, pytest, Logging & Metrics, API Testing, Linux

Workflow & PM: Agile/Scrum, Cross-functional Collaboration, Technical Design Docs, Stakeholder Communication